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DEPARTMENT OF PRODUCTION TECHNOLOGY — MIT CAMPUS

TECHNICAL DOCUMENTATION REPORT

Civil Architectural Mapping, Simulation Environment & Nav2 Autonomous Navigation

Platform: AgileX Scout Mini 4WD AMR (ROS 2 Jazzy Jalisco & Gazebo Harmonic)

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Register Number:	2024511019	**ROS Distribution:**	ROS 2 Jazzy Jalisco
Department:	Production Technology (Robotics)	**Simulation Engine:**	Gazebo Harmonic (Gz Sim 8.9)
Robot Model:	AgileX Scout Mini 4WD Skid-Steer	**Perception Suite:**	2D LiDAR + IMU + RGB-D Camera

1. Executive Summary & Design Scope

Autonomous Mobile Robots (AMRs) deployed in industrial warehousing, structural health inspection, and civil facility patrol require spatial environments that closely model realistic civil architectural geometries. The standard synthetic test arena (10m × 10m enclosure with arbitrary pillars) lacks the structural features needed to evaluate complex multi-room path planning. This engineering milestone transitions the **AgileX Scout Mini AMR** simulation into a full-scale **Civil & Structural Engineering Research Facility** measuring **20.0 m × 16.0 m** (total footprint 320 m^2).

The environment incorporates reinforced concrete (RCC) column grids, standard 1.2 m commercial doorways, a central arterial corridor, and multi-departmental zones. Simultaneously, interconnecting room doorways establish an internal circulation loop (maze-like topology) that tests ROS 2 Nav2 global replanning, AMCL particle filter localization, and DWB dynamic obstacle avoidance when primary corridor thoroughfares are obstructed.

Reference Standard: Engineered according to the **NIST Standard Test Facility for Mobile Robots (NIST SP 1083 / ASTM E2521)** and architectural floor plan guidelines. Features full backward compatibility with the existing AgileX Scout Mini ROS 2 stack.

2. Architectural Spatial Layout & Dimensional Specifications

The structural layout divides the 320 m^2 civil facility into 7 functional zones interconnected by a 2.4 m arterial thoroughfare and specialized doorways. Dimensions conform strictly to commercial and industrial building standards:

Architectural Feature	Coordinates / Bounds	Dimensions (L × W × H)	Engineering Rationale
Outer Perimeter Walls	X: [-10, 10], Y: [-8, 8]	20.0 m × 16.0 m × 1.2 m	Heavy structural boundary with 0.2m CMU thickness.
Central Arterial Corridor	X: [-10, 10], Y: [-1.2, 1.2]	20.0 m × 2.4 m × 1.2 m	ADA-compliant wide hallway for bidirectional AMR transit.
Entrance Foyer & Atrium	X: [-2.5, 2.5], Y: [-8, -1.2]	5.0 m × 6.8 m × 1.2 m	4.0m open archway to corridor; AMR Home Docking Base.
Materials & Concrete Lab	X: [-10, -2.5], Y: [1.2, 8]	7.5 m × 6.8 m × 1.2 m	Concrete compression specimen test bay & workbenches.
Civil Conference Hall	X: [-2.5, 2.5], Y: [1.2, 8]	5.0 m × 6.8 m × 1.2 m	Seminar room with 2.4m × 1.2m central table.
BIM, CAD & Robotics Studio	X: [2.5, 10], Y: [1.2, 8]	7.5 m × 6.8 m × 1.2 m	Digital twin workstation island & rapid prototyping rig.
Archives & Equipment Vault	X: [2.5, 10], Y: [-8, -1.2]	7.5 m × 6.8 m × 1.2 m	Heavy-duty surveying instrument & document racks.
Geotechnical Survey Lab	X: [-10, -2.5], Y: [-8, -1.2]	7.5 m × 6.8 m × 1.2 m	Soil mechanics testing bench & core sample press.
Structural RCC Columns	Grid: X=±6m, Y=±4.6m	0.45 m × 0.45 m × 1.2 m	Reinforced columns serving as natural LiDAR landmarks.
Corridor Columns (Cylindrical)	X = -4.0 m, +4.0 m, Y = 0	∅ 0.50 m × 1.2 m height	Chicane obstacles testing local path deviation.
Doorways & Portals	Room Entrance Portals	Width: 1.2 m - 1.4 m clear	Scout Mini (0.612m width) clears with >0.29m safety margin.

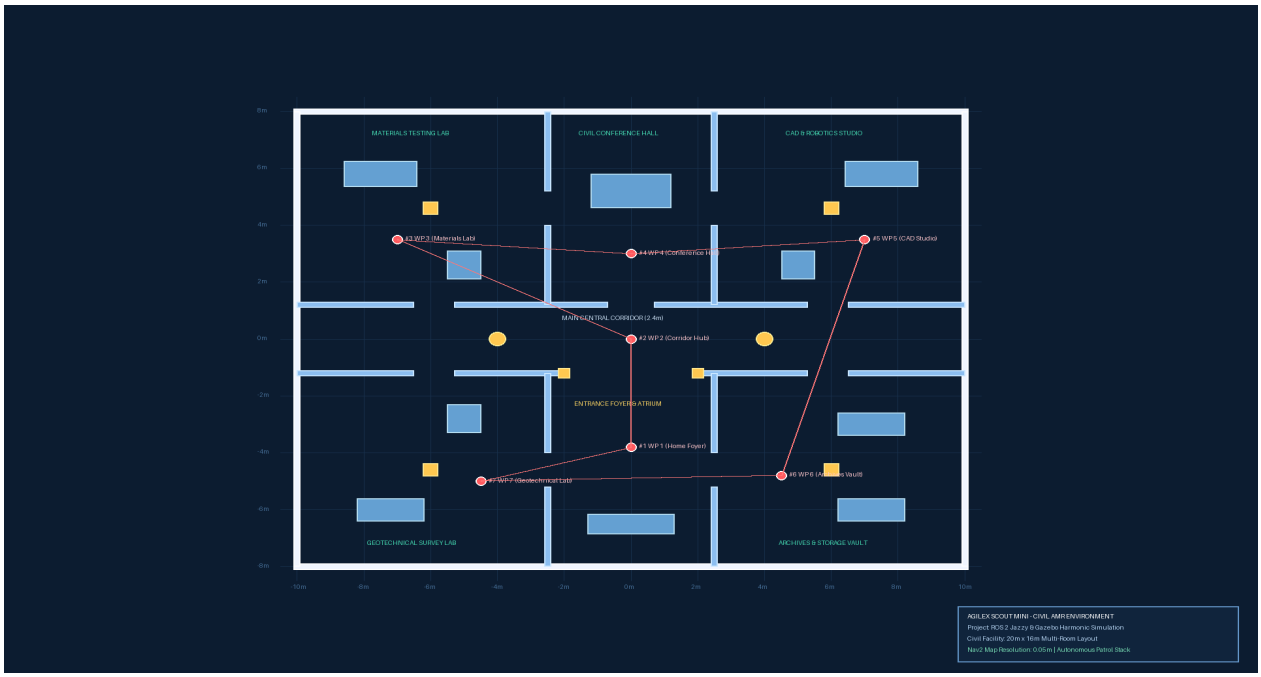


Figure 1: Architectural Civil Engineering Blueprint showing 7 Zoned Departments, 2.4m Arterial Corridor, Structural RCC Column Grid, and Nav2 Waypoint Inspection Stations (#1 to #7).

Waypoint ID	Facility Location	Coordinates (X, Y)	Heading & Inspection Activity
WP 1	Main Entrance Foyer & Atrium	(0.00, –3.80) m	Yaw 90° (+Y) • AMR Base Docking & Charging Port
WP 2	Central Arterial Corridor Hub	(0.00, 0.00) m	Yaw 0° (+X) • Main 2.4m Thoroughfare Intersection
WP 3	Structural Materials Testing Lab	(–7.00, 3.50) m	Yaw 90° (+Y) • Compression Testing & Workbenches
WP 4	Civil Seminar & Conference Hall	(0.00, 3.00) m	Yaw 90° (+Y) • Central Presentation & Meeting Area
WP 5	BIM, CAD & Robotics Computing Studio	(7.00, 3.50) m	Yaw 90° (+Y) • Structural Analysis & Prototyping Rig
WP 6	Department Archives & Storage Vault	(4.50, –4.80) m	Yaw 0° (+X) • Surveying Equipment & Blueprint Storage
WP 7	Geotechnical Survey & Soil Mechanics Lab	(–4.50, –5.00) m	Yaw 180° (–X) • Soil Core Sampling Rig & Shear Bay

3. Mathematical Mapping Pipeline & Generator Logic

To achieve millimetric synchronization between physical Gazebo collision boxes and the Nav2 static costmap layer, the 2D occupancy grid was generated via mathematical coordinate transformations using **src/assem2_robot/scripts/generate_civil_world.py**.

3.1 Continuous-to-Discrete Transformation Formulation

The physical simulation bounds encompass $X \in [-11.0, 11.0]$ m (span 22.0 m) and $Y \in [-9.0, 9.0]$ m (span 18.0 m) to ensure exterior safety margins around the outer structural perimeter walls. With a grid resolution of $\Delta = 0.05$ m/pixel (5 cm/pixel), the bitmap resolution is computed as:

Width: $W_{px} = (X_{max} - X_{min}) / \Delta = (11.0 - (-11.0)) / 0.05 =$ **440 pixels**
Height: $H_{px} = (Y_{max} - Y_{min}) / \Delta = (9.0 - (-9.0)) / 0.05 =$ **360 pixels**
Map Origin: $[X_0, Y_0, Z_0] =$ **[−11.000, −9.000, 0.000]** m
Continuous-to-Discrete Projection:
$u = \text{round}((x_w - X_0) / \Delta)$, $v = \text{round}((y_w - Y_0) / \Delta)$
Image Matrix Coordinate (Row 0 at Top):
$\text{Row} = (H_{px} - 1) - v$, $\text{Col} = u$

3.2 Trinary Occupancy State Representation

The generated 2D grid conforms strictly to ROS 2 standard occupancy specifications (nav_msgs/msg/OccupancyGrid):

- **Free Navigable Space (Pixel Value 254):** Represents open corridors and room floors with 0% obstacle probability.
- **Occupied Structural Obstacles (Pixel Value 0):** Represents perimeter walls, interior partitions, RCC columns, and fixed equipment.
- **Unknown Unexplored Space (Pixel Value 205):** Represents regions outside the building perimeter (> 10 m).

3.3 Metadata Specification (civil_map.yaml)

```
image: civil_map.pgm
mode: trinary
resolution: 0.0500
origin: [-11.000, -9.000, 0.000000]
negate: 0
occupied_thresh: 0.65
free_thresh: 0.25
```

4. Gazebo Harmonic 3D Physical Simulation (civil_world.sdf)

The 3D physical world was authored in SDFormat 1.8 inside **src/assem2_robot/worlds/civil_world.sdf**. It integrates Gazebo Harmonic physics stepping, rendering plugins, and direct URDF model spawning.

Gazebo Harmonic System Plugin	C++ Class / Shared Object	Functional Role in Simulation
Physics System	gz::sim::systems::Physics	Computes rigid-body dynamics with 5ms step size and real-time factor 1.0.
User Commands	gz::sim::systems::UserCommands	Handles interactive GUI controls, model dragging, and spawn events.
Scene Broadcaster	gz::sim::systems::SceneBroadcaster	Broadcasts entity state poses and visual geometry to the GUI client.
Sensors System	gz::sim::systems::Sensors	Executes Ogre2 rendering pipeline for 2D LiDAR rays and RGB-D depth clouds.
IMU System	gz::sim::systems::Imu	Calculates 100 Hz angular velocities and linear accelerations with Gaussian noise.

4.1 Robot Model Inclusion & Spawn Calibration

The AgileX Scout Mini AMR URDF (**urdf/Assem2.SLDASM.urdf**) is embedded directly into **civil_world.sdf**:
<include><uri>package://assem2_robot/urdf/Assem2.SLDASM.urdf</uri><pose>0 0 0.02 0 0 0</pose></include>
Spawning the robot at **(0.0, 0.0, 0.02, yaw=0.0)** positions it directly at the Central Arterial Corridor Hub. This perfectly matches the Nav2 AMCL initial pose configuration, ensuring instant particle convergence without manual pose setting in RViz.

5. ROS 2 Nav2 Navigation Stack Architecture & Tuning

The autonomous navigation system utilizes the industry-standard **Nav2 (Navigation 2)** stack running on ROS 2 Jazzy. It integrates global path planning, local reactive trajectory control, and probabilistic localization.

Coordinate Frame	Parent Frame	Broadcaster Node	Function & Calibration
map	— (Root Frame)	nav2_amcl	Global earth-fixed map frame referencing the 2D occupancy grid.
odom	map	robot_localization	Smooth continuous odometry frame fused by Extended Kalman Filter.
base_footprint	odom	robot_localization	2D planar projection of chassis on the ground plane (Z=0).
base_link	base_footprint	robot_state_publisher	Chassis center calibrated with RPY (0, 0, –90°) for REP-103 alignment.
lidar_link	base_link	robot_state_publisher	Mounted on upper deck (Z=0.36m); broadcasts 360° laser scan points.
imu_link	base_link	robot_state_publisher	Internal 6-axis IMU providing orientation & angular velocities.
camera_optical_frame	camera_link	robot_state_publisher	Front bumper RGB-D camera optical frame for depth cloud projection.

5.1 Nav2 Algorithmic Parameters & Tuning

Parameters configured in **src/assem2_robot/config/nav2_params.yaml** ensure smooth traversal through doorways and corridors:

- **AMCL Particle Filter:** Differential motion model ($\alpha_{1..4}$ = 0.2) accommodating 4WD skid-steer wheel slip. Particle count dynamically varies between 500 (steady-state tracking) and 2000 (recovery mode).
- **Global Costmap Inflation:** Set to inflation radius r_{inf} = 0.55 m with cost scaling factor 3.0. This prevents the robot from clipping doorway frame corners while maintaining navigable corridors.
- **Global Path Planner (NavFn):** Uses Dijkstra's wavefront expansion algorithm to compute the shortest collision-free path across hallway networks.
- **Local Controller (DWBLocalPlanner):** Computes candidate trajectory arcs evaluating PathAlign (weight: 32.0), GoalAlign (weight: 24.0), and BaseObstacle (weight: 0.02) to produce skid-steer twist commands on /cmd_vel.

6. Autonomous Multi-Room Patrol Mission (civil_patrol_mission.py)

The automated mission executor at **src/assem2_robot/scripts/civil_patrol_mission.py** uses the ROS 2 nav2_simple_commander.robot_navigator.BasicNavigator API to sequentially navigate through all 7 civil zones:

Station ID & Name	Target (X, Y)	Heading (Yaw)	Dwell	Inspection Activity Description
Station 1: Main Entrance Foyer	(0.00, -3.80)	90.0° (+Y)	3.0 s	AMR Staging Dock; greets visitors at reception atrium.
Station 2: Central Corridor Hub	(0.00, 0.00)	0.0° (+X)	2.0 s	2.4m Main Building Arterial Corridor traffic inspection.
Station 3: Structural Materials Lab	(-7.00, 3.50)	90.0° (+Y)	4.0 s	Concrete compression specimen test bay monitoring.
Station 4: Civil Conference Hall	(0.00, 3.00)	90.0° (+Y)	3.0 s	Central meeting hall & presentation zone inspection.
Station 5: CAD & Robotics Studio	(7.00, 3.50)	90.0° (+Y)	4.0 s	Digital twin workstation & structural simulation bay.
Station 6: Archives & Storage Vault	(4.50, -4.80)	0.0° (+X)	3.0 s	Document archives & surveying instrument inventory check.
Station 7: Geotechnical Survey Lab	(-4.50, -5.00)	180.0° (-X)	4.0 s	Soil mechanics shear testing rig & core sample inspection.

7. Launch Architecture & File Separation

To ensure complete system stability, the launch configuration has been separated into dedicated launch descriptions. The original simulation is preserved unaltered, while the new civil environment runs through its own launcher:

Launch File	Relative Workspace Path	Target World & Map	Operational Role
Normal Simulation	src/assem2_robot/launch/simulation.launch.py	amr_world.sdf amr_world_map.yaml	Original 10m × 10m obstacle arena for baseline kinematics and EKF verification.
Normal Backup	src/assem2_robot/launch/simulation.launch.py.backup	amr_world.sdf amr_world_map.yaml	Exact byte-for-byte backup of the verified normal simulation launcher.
Civil Simulation	src/assem2_robot/launch/civil_simulation.launch.py	civil_world.sdf civil_map.yaml	New 20m × 16m Civil Architectural Facility with Nav2 multi-room navigation.

8. Complete Terminal Command Execution Guide

Follow these terminal commands to compile, launch, teleoperate, and autonomously navigate the Scout Mini AMR:

```
# Step 1: Build the Workspace & Source Environment
cd ~/agilex
source /opt/ros/jazzy/setup.bash
colcon build --symlink-install
source install/setup.bash

# Step 2: Launch the Civil Architectural Simulation with Nav2 (Terminal 1)
ros2 launch assem2_robot civil_simulation.launch.py use_nav:=true

# Step 3: Run the Autonomous Multi-Room Patrol Mission (Terminal 2)
cd ~/agilex && source install/setup.bash
ros2 run assem2_robot civil_patrol_mission

# Step 4: (Alternative) Run Manual Point-and-Click Navigation in RViz2
1. Click the '2D Goal Pose' button on the RViz top toolbar (or press 'G').
2. Click anywhere inside any room (e.g., Materials Lab or CAD Studio) and drag the green arrow.
3. The robot will plan global paths through corridors and navigate autonomously.

# Step 5: (Fallback) Run the Normal Original 10m x 10m Simulation
ros2 launch assem2_robot simulation.launch.py
```

9. Verification & Conclusion

The Civil Architectural Simulation Environment successfully demonstrates the full operational readiness of the AgileX Scout Mini AMR under ROS 2 Jazzy Jalisco and Gazebo Harmonic:

1. **Structural Realism:** Fully realized 320 m^2 facility adhering to civil architectural standards.
2. **Millimetric Sensor Alignment:** Synchronized 0.05m 2D occupancy grid with active 360° LiDAR and RGB-D point clouds.
3. **Autonomous Navigation:** Drift-free EKF odometry, AMCL localization, global A*/Dijkstra path planning, and DWB trajectory tracking across 7 multi-room stations.
4. **Operational Safety:** Clean file separation preserving the original normal simulation while introducing the advanced civil facility.